

PHYS 253: Thermal Physics

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August 31st, 2026

Abstract

NOT FINISHED

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1 Introduction

Definition 1 (Thermodynamics). *Thermodynamics, derived from the ancient Greek words *therme* (heat) and *dynamis* (power), is the branch of physics that deals with the relationships between heat, work, temperature, and energy.*

Remark 1. *Thermodynamics does not give microscopic details of the system¹, but rather describes the macroscopic properties and behaviors of systems in terms of a few state variables.*

2 Energy Conservation

3 Thermal Equilibrium

4 Entropy

A Appendix

B Useful Links

¹Classical thermodynamics cannot describe microscopic systems directly, but modern extensions like stochastic thermodynamics and statistical mechanics can.